Autonomous AI agents discover an accessory component of site-specific recombination systems controlling mobile DNA integration and excision

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Abstract

Protein sequence resources contain many conserved families whose biological roles remain unresolved. Assigning functions to these families requires both appropriate statistical comparisons and retrieval of evidence across inconsistent annotations. Here we describe an autonomous genome-mining campaign coordinated by Luria that surveyed 618,638,921 proteins through five complementary search strategies. None recovered a new programmable system under the campaign's criteria. An integrase-neighborhood screen nevertheless nominated DUF2924 (PF11149), a small, predominantly basic protein family. Because the original enrichment was sensitive to background composition, the analysis was reformulated around all detected recombinase loci. DUF2924 occurred within five genes of 14,088 of 524,414 serine recombinases and 665 of 1,403,550 tyrosine integrases, corresponding to an odds ratio of 58.2. Its partners were predominantly large serine recombinases. Among 168 immediately adjacent pairs with coordinates, all were on the same strand, and gene overlaps were common. Junction-spanning RNA-seq reads supported cotranscription at a representative locus, while enrichment within predicted proviruses linked the family to mobile genetic elements. Sequence and structural analyses suggested possible nucleic-acid recognition but did not establish binding or recombination directionality. We propose the name OPAL, operon-coupled accessory of large serine recombinases, as a provisional designation for this genomic association. These findings identify a candidate accessory family and illustrate how an auditable autonomous workflow can revise its statistical framing while preserving negative and inconclusive results.

In brief An autonomous genome-mining campaign that did not recover a new programmable system identified DUF2924 as a candidate accessory domain of large serine recombinases. Population-wide association tests, conserved gene organization and junction-spanning RNA-seq reads support an operon-level relationship; its biochemical role remains to be determined.

Introduction

Genome and metagenome sequencing have expanded the accessible diversity of microbial proteins, creating opportunities to discover molecular activities and genome-engineering systems. Resources such as the Genome Taxonomy Database (GTDB), RefSeq, Pfam and InterPro provide complementary frameworks for organizing this diversity by taxonomy, sequence and domain architecture [1, 2, 3, 4]. Discoveries such as bridge RNA-guided recombination demonstrate how investigation of mobile genetic elements can reveal previously unrecognized mechanisms of sequence recognition and DNA rearrangement [5]. Nevertheless, family-level annotation frequently precedes experimental characterization. In the Pfam model collection used for the present campaign, 6,558 of 30,134 models carried unknown-function labels; these are counts from the campaign's database snapshot, rather than a statement about the current public release.

Conserved genomic neighborhoods offer one route from sequence families to biological hypotheses. Gene proximity, shared orientation and recurrent domain combinations can indicate functional coupling, although each provides evidence at a different level of specificity [6, 7, 8]. Enrichment screens are especially sensitive to their comparison populations: differences in protein length, taxonomic representation or mobile-element content can produce an association without identifying its cause. A low adjusted p value addresses multiplicity under a specified model; it does not correct an inappropriate background [9]. Literature retrieval creates a related problem when the same family is described under an accession, an older name or a broader domain classification. Both issues require revision of the question as well as execution of the analysis.

Automated scientific systems have progressed from closed experimental workflows to language-model agents that plan analyses, use external tools and coordinate iterative investigations [10, 11, 12]. Such systems make genome mining a tractable setting for studying scientific autonomy: datasets are large, analytical steps are diverse, and candidate findings require repeated checks against alternative explanations. A central question is whether an agentic workflow can recognize that its initial analysis does not justify a claim and adopt a more informative comparison. An auditable record is useful here because successful candidates alone do not reveal discarded hypotheses, inappropriate controls or unsuccessful search branches.

Site-specific recombinases provide a biologically consequential test case. Tyrosine and serine recombinases catalyze DNA rearrangement through distinct mechanisms. Large serine recombinases (LSRs), including Bxb1 and ϕC31 integrases, combine an N-terminal catalytic domain with C-terminal regions involved in DNA recognition and synaptic control [13]. Their ability to integrate substantial DNA cargos has supported genome-engineering approaches that combine integrases with programmable targeting systems [14, 15]. Native LSRs recognize cognate attachment sites; arbitrary target programmability should therefore be distinguished from site specificity or programmability supplied by an engineered targeting component.

In several characterized systems, recombination directionality factors (RDFs) regulate integration and excision through interactions with their cognate integrases [16, 17, 18, 19, 20]. These factors are heterogeneous in sequence, size and genomic placement. Recent work has combined structural prediction with experimental validation to identify additional LSR–RDF pairs, so the known repertoire extends beyond the five historical examples considered in the initial campaign [21]. Identifying further accessory families requires distinguishing a conserved genomic association from experimentally demonstrated directionality control.

Here we report a Luria-coordinated campaign that searched 618,638,921 proteins through five complementary strategies. The campaign did not recover a new programmable system under its search criteria, but it nominated DUF2924 (PF11149). Reformulating the analysis around all detected recombinase loci revealed a strong association with serine recombinases, supported by immediate adjacency, conserved orientation and mobile-element context. We propose OPAL as a provisional name for this candidate accessory domain. The results establish a genomic and transcriptional relationship while leaving its molecular activity open to experimental testing.

Results

An auditable genome-mining campaign with Luria

Luria coordinated language-model agents through a shared, append-only ledger recording analytical rounds, declared beliefs, hypotheses, prospective falsification criteria, role dispatches and stopping decisions (Figure 1A). An audi-

tor checked reported quantities against the recorded evidence, while a blinded experimental role reproduced selected measurements without their motivating interpretation. A drafting step assembled the report from this record, and a manuscript gate tracked missing evidence, unsupported statements, figure failures and structural failures. These checks document the workflow; they do not independently establish the validity of every recorded statistic.

Across 13 runtime turns, the campaign recorded 25 campaigns, 93 rounds, 18 declared beliefs, 45 theories, six refuted theories, seven prospective falsifiers, 76 role dispatches and 27 stops (Figure 1A,C). Of 11 defined roles, two were dispatched: the auditor 63 times and the blinded experimental role 13 times (Figure S1A). The reported token total was 1,365 million, including 10 million generated tokens. External direction was supplied at the beginning of each runtime turn. Accordingly, autonomy here refers to execution and revision within these directed intervals, rather than a fully unsupervised study.

A negative primary search nominates DUF2924 as a secondary lead

The campaign used a collection reported as 618,638,921 proteins, drawing on GTDB representative proteomes and viral resources, including RefSeq [1, 2]. Five complementary search strategies examined gene context, noncoding RNA, domain architecture, predicted structure and DNA repeat arrays (Figure 1B). None yielded a new programmable system meeting the campaign's criteria. The context-first branch recovered host Holliday-junction resolvases; the structure-first branch examined eight orphan predictions without assigning a relevant match; and the repeat-array branch promoted no candidate in the sampled 500 bacterial and 5,000 phage genomes. These outcomes delimit the searches performed and do not demonstrate that the underlying sequence collection lacks undiscovered programmable systems.

A secondary Pfam screen compared proteins near integrases with a background sample [3, 22]. Of 30,134 models, 4,857 had sufficient counts for testing, and 32 reached a reported Benjamini–Hochberg false discovery rate of $q < 0.05$ (Figure 1D; Figure S1B) [9]. Thirty-one had an alternative explanation: five recapitulated the selection criterion, 23 represented familiar mobile-element-associated domains, two were the N- and C-terminal domains of RuvA, and one was DUF4158. RuvA enrichment was consistent with host *ruvAB* loci, because 31 of 33 loci also contained adjacent *ruvB*. DUF4158-containing proteins had a median length of 984 residues, and 10 of 13 also carried a Tn3 transposase domain. DUF2924 remained unexplained, with 32 case proteins and no background hit (reported one-sided Fisher's exact $p = 2.3 \times 10^{-10}$, $q = 9.2 \times 10^{-8}$).

Several checks limited interpretation of this screen. Eleven housekeeping domains were depleted, with a median odds ratio of 0.566, but case proteins were shorter than background proteins (median 211 versus 273 residues; 36% versus 23% classified as short). Only 5,757 contigs were shared between samples, representing 5.9% of background contigs (Figure S1C). Restricting analysis to proteins longer than 400 residues addressed some families but left no eligible DUF2924 observations. This restriction therefore could not resolve the length confound for DUF2924. A separate example illustrated the effect of case definition: *pyrG* enrichment fell from 5.67-fold to an odds ratio of 0.26 after 20,585 host Xer-associated sites were removed from 100,000 case loci, whereas the *guaA* signal strengthened from 3.44 to 4.28.

Reformulating the denominator reveals a serine-recombinase association

The DUF2924 lead was initially deprioritized after an accession-based literature search failed to recover a functional explanation. The sequence of recorded entries below is verbatim from the ledger and from the agents' own reasoning traces, reproduced here with timestamps and attribution so that the recognition process itself is auditable, independent of the quantitative claims it led to.

The lead first appears as one line among several unresolved candidates from the Pfam screen described above:

> DUF2924（PF11149.14）case 32 / bg 0，q=9.2e-8。功能未知域，待 swissprot/nr + prior art 查。

Ledger entry, 2026-10-01 20:54 UTC.

Translation: DUF2924 (PF11149.14), case 32 / bg 0, $q = 9.2 \times 10^{-8}$. Function unknown; pending a SwissProt/nr search and a prior-art check.

> ## 还没判的两条
> “‘ RuvA_N 7.0× (case 35/bg 5) RuvA_C 6.6× (33/5) q≈1.5e-4 / 4.4e-4 DUF2924 ∞ (case 32/bg 0) q=9.2e-8

Supervising instance, continuation planning, 2026-10-01 20:54 UTC.

Translation: Two undecided leads. RuvA_N 7.0× (case 35/bg 5); RuvA_C 6.6× (33/5), $q \approx 1.5 \times 10^{-4}/4.4 \times 10^{-4}$. *DUF2924: infinite enrichment (case 32/bg 0),* $q = 9.2 \times 10^{-8}$.

A literature search by domain identifier found nothing, and a direct sequence search confirmed the domain had no informative match in SwissProt:

> So DUF2924 has NO meaningful swissprot hit — it’s a genuinely uncharacterized domain (DUF = Domain of Unknown Function).

Worker agent, 2026-10-01 21:07 UTC.

Six minutes later, the same agent restated the conclusion while comparing DUF2924 against the other open lead:

> DUF2924 is the only genuinely ”unannotated” lead (32/0, no function known anywhere), but it’s a DUF (unknown function), not a ”novel programmable enzyme”.

Worker agent, 2026-10-01 21:13 UTC.

Attention moved to RuvA, which subsequently resolved into a host *ruvAB* operon (Figure 1D) rather than cargo. The DUF2924 lead reopened after sequence clustering, described above, placed it beside large-serine-recombinase domains. Before any measurement, the agent recorded three competing readings of that observation:

> But actually, the campaign is more of a ”competing hypotheses” campaign: - Hypothesis A: DUF2924 is a serine-recombinase-specific accessory (the 58.5× specificity is real and new). - Hypothesis B: DUF2924 is a generic MGE accessory, not serine-specific (the 58.5× is a sampling artifact). - Hypothesis C: DUF2924 is a host-modification reader (RAMA DNA-binding), associated with recombinases incidentally.
> The reverse rate (main task) discriminates A vs B: if serine loci have a high DUF2924 rate and tyrosine loci low, A is supported; if both low, DUF2924 is a rare accessory (B).

Worker agent, 2026-10-02 04:02 UTC.

Hypothesis C motivated a separate check: a dispatched literature-verification task confirmed the exact published wording relating DUF2924 to RAMA, and corrected an internal paraphrase that had overstated it:

> ”DUF2924 = RAMA” is a PARTIAL match, not a strict identity. The literature says ”RAMA DNA-binding domain (partly matching Pfam DUF2924)”. So the correct statement is ”DUF2924 partly overlaps/partly matches the RAMA domain”, NOT ”DUF2924 IS RAMA”.

Worker agent, reporting a literature-verification subagent, 2026-10-02 04:10 UTC.

The reverse-rate measurement itself, reported quantitatively above, was first returned as follows:

> The reverse rate is a STRONG positive result: Serine recombinase with DUF2924 ±5: 14,088 / 524,414 = 2.69% (1 in 37 serine recombinase). Tyrosine integrase with DUF2924 ±5: 665 / 1,403,550 = 0.0474% (1 in 2,110 tyrosine integrase). Odds ratio: 58.24×. Fisher p = 0. This is a STRONG confirmation of the forward 58.5× specificity, and it matches almost exactly (58.24× vs 58.5× forward).

Worker agent, 2026-10-02 04:35 UTC.

A secondary measurement in the same campaign, on methyltransferase co-occurrence, produced the only moment of explicit surprise recorded anywhere in the run, together with the self-check that followed it in the same message:

> Interesting! The methylation control result is UNEXPECTED: […] This is the OPPOSITE of what the housekeeper expected. […] Wait, let me reconsider. This is surprising. Let me think about whether this is a real signal or an artifact.

Worker agent, 2026-10-02 04:56 UTC.

This passage is reproduced to document how the association was recognized, not as evidence for the association itself; the quantitative support for each claim it reaches is reported separately, with its own controls, in the paragraphs that follow.

The analysis was then reversed: among all detected serine recombinase and tyrosine integrase loci, what fraction carried DUF2924 within five genes? DUF2924 was present at 14,088 of 524,414 serine recombinase loci (2.686%) and 665 of 1,403,550 tyrosine integrase loci (0.0474%), giving an odds ratio of 58.2 and a prevalence ratio of approximately 56.7 (Figure 1D; Figure S2A). The recorded Fisher calculation underflowed floating-point precision; this is a numerical limitation, not a literal probability of zero. Among serine-associated cases, 91.0% were separated by exactly one gene, whereas the tyrosine-associated distribution was more dispersed. The population-wide comparison removes dependence on the original sampled background but can still be influenced by taxonomy, sequence redundancy and annotation sensitivity.

A comparison restricted to shared contigs also supported the association: 30 of 8,491 case proteins carried DUF2924, compared with none of 7,558 background proteins (reported $p = 5.1 \times 10^{-9}$; Figure S2C). The campaign assigned the reference loci to at least 35 genera, although taxonomic resolution varied across assemblies. These observations support a broadly distributed association rather than proof of independent evolutionary origins.

Methyltransferases occurred near 36 of 183 DUF2924 loci (19.7%) and 12 of 200 integrase-adjacent orphan controls (6.0%; prevalence ratio 3.28; reported $p = 8.1 \times 10^{-5}$). This prompted consideration of RAMA, the restriction enzyme–adenine methylase-associated domain, which partially overlaps the DUF2924 annotation in prior literature [23, 24]. GANTC motif density near DUF2924 exceeded the comparison set by only 0.13 standard deviations, while the GATC control shifted by −0.07 standard deviations. This analysis offered little support for the proposed sequence-density signature. It does not measure DNA methylation or directly test recognition of CcrM-methylated DNA, and therefore cannot exclude methylation-dependent binding.

Most sampled partners have large serine recombinase architectures

Domain classification distinguished small serine resolvases from LSRs, whose additional DNA-binding regions support site-specific integration and excision [13]. In the campaign's neighborhood classification, 166 of 183 DUF2924 loci were associated with an LSR, ten with a Resolvase-only protein and seven with neither category. Of the 166 LSR-associated loci, 159 contained proteins with both Resolvase and Recombinase domains, including 64 that also carried a zinc ribbon (Figure 2A,B). These neighborhood-level counts differ from the stricter immediately adjacent pair set used for gene-spacing analysis.

A tree of 205 recombinase catalytic domains showed strong grouping by DUF2924 association: adjacent tips shared a label at 200 of 204 positions. Restricting the comparison to LSRs gave 171 of 175 same-label adjacencies (reported permutation $z = 11.9$, $p = 2 \times 10^{-4}$). The associated set had a median pairwise identity of 69%. This pattern is consistent with concentration in a sampled LSR lineage, but the control collection contained only 18 LSRs selected separately. Moreover, tip adjacency depends partly on how tree branches are ordered. A broadly representative control set and topology-based association test would be needed to establish specificity to a distinct evolutionary subfamily.

Position-resolved annotations reinforced the local association. Across the 183 neighborhoods, Resolvase occurred at the two positions immediately flanking DUF2924 in 44% and 47% of loci; DUF3489 occurred at 11% on either side, and no other domain exceeded 4% at a single position (Figure 2C). Methyltransferase and restriction-modification annotations were distributed farther from DUF2924 (Figure 2H). The immediate recombinase association and broader methyltransferase association therefore describe different spatial relationships.

DUF2924 is predominantly a small basic protein family

Of 183 proteins in the reference collection, 181 consisted predominantly of DUF2924, while two longer proteins, 469 and 719 residues, also carried recombinase domains. The full collection had a median length of 157 residues (Figure 2G). DUF2924 proteins had a median net charge of +6.4 per 100 residues, compared with +1.4 in length-matched neighborhood proteins (reported Mann–Whitney $p = 3 \times 10^{-31}$); 181 of 183 had positive net charge. The 24 predicted structures also displayed predominantly positive exposed surfaces. In one model, the most basic view contained 51% basic and 8% acidic area, compared with 29% and 22% in the least basic view (Figure 2D,E). These features motivate nucleic-acid-binding assays but do not identify a substrate.

Foldseek searches did not assign a confident structural match to any of the 24 models at $E < 10^{-3}$; the best reported match had $E = 0.074$ (Figure 2F) [25]. Three models had weak matches to MPND, a RAMA-containing protein studied in DNA recognition [26, 27]. Together with the partial sequence annotation, these matches motivate comparison with RAMA-related proteins [23]. They do not establish a common fold, modified-base specificity or functional equivalence to MPND. Published descriptions of RAMA/DUF2924 and its genomic association in *Wolbachia* also preclude treating the domain as previously unmentioned [24, 28, 29].

Conservation was strongest near the C terminus. Among 134 alignment columns occupied in more than half of the 183 sequences, columns 171–190 had a mean conservation score of 0.723, compared with 0.441 elsewhere. The segment included the consensus `LSA.A..ITGT.WNG..FF`; its conserved glycines and terminal aromatic residues provide candidates for mutagenesis (Figure 2I; Figure S3B,C). These are alignment-column coordinates, rather than universal residue numbers. The segment also had greater information content in the Pfam model (0.99 versus 0.59 bits). RAMA annotations and recombinase proximity occurred across the sampled DUF2924 tree, although the supplied summaries use slightly different annotation counts (Figure S3A).

Gene organization and RNA-seq support operon-level coupling

Among 15,266 DUF2924 proteins detected in the larger search, 13,494 could be paired with a nearby serine recombinase. Of these, 12,790 (94.8%) placed the recombinase immediately adjacent (Figure 3C). This DUF2924-centered fraction has a different denominator from the 91.0% recombinase-centered fraction reported above.

In the coordinate-resolved collection, 168 immediately adjacent pairs contained a Resolvase-bearing partner, and all 168 were on the same strand. The stored intergenic coordinate difference had a median of −3 bp, with 89 pairs at that value (Figure 3D). Inspection of the supplied inclusive coordinates shows that a difference of −3 corresponds to a four-nucleotide CDS overlap. The representative sequence is `ATGA`, containing an `ATG` start and a `TGA` stop. The two codons share the dinucleotide `TG`; they do not share three bases. Such arrangements are consistent with translational coupling, which can arise through termination–reinitiation or other mechanisms [8, 30].

The coordinate convention also affects the overlap count. There are 127 strictly negative stored differences and 17 zero differences. With inclusive coordinates, zero difference denotes a one-nucleotide overlap, so 144 of 168 pairs (85.7%) overlap by at least one nucleotide. Under this convention, the stored overlap-phase counts correspond to true overlap lengths in two residue classes modulo three. Their distribution is not uniform, but start/stop codon constraints make a uniform phase distribution an insufficient mechanistic null. The overlap pattern supports constrained gene organization; it does not by itself demonstrate ribosome reinitiation. The original figure retains the campaign's coordinate convention and is explicitly annotated below to prevent conflation with nucleotide overlap length.

Two public RNA-seq runs, `SRR12463562` and `SRR12463566`, provided independent sequencing observations at the orthologous locus in *Rhodobacter capsulatus* SB1003. A 152-residue DUF2924 protein and a 441-residue LSR were encoded by overlapping genes on the minus strand. Eighteen of 41 and five of 14 reads overlapping the junction window spanned it completely (Figure 3E). These reads support cotranscription at this locus. Coverage was limited, with median depths of 11 and eight over DUF2924 and three and one over the recombinase. The observations do not establish a shared promoter, universal cotranscription across the family or translational dependence between the two cistrons.

Partner architecture did not show an obvious reduction associated with DUF2924. In the comparison of 162 associated and 19 control recombinases, median lengths were 521 and 551 residues (Mann–Whitney $p = 0.30$), and zinc-ribbon frequencies were 41% and 53% (Fisher's exact $p = 0.34$; Figure 3F). These nonsignificant differences are not evidence of equivalence, particularly given the smaller control group.

Provirus localization supports a mobile-element association

The geNomad analysis identified 87 predicted proviruses across the 181 reference contigs [31]. Twenty-nine of 183 DUF2924 loci (15.8%) lay within these regions, which occupied 1.80% of the analyzed sequence length, giving a descriptive observed-to-coverage ratio of 8.8 (Figure 3G). A binomial calculation using sequence coverage as the placement probability gives $p \approx 7.0 \times 10^{-19}$, but assumes independent, uniformly positioned loci. Because this null ignores gene density, taxonomy and element-associated sampling, the effect size is more informative than its nominal significance. Loci outside predicted proviruses may occur in other mobile elements or unrecognized viral regions; the analysis does not classify all such loci.

A representative locus in *Desulfosarcina ovata* subsp. *ovata* oXyS1 lay within an 89.9-kb predicted provirus on a 7.63-Mb chromosome, near a CI-like repressor (Figure 3B). Across the 29 provirus-associated loci, the median distance to the nearer predicted boundary was 0.114 of element length. Twenty-two lay within the outer 20% at either end (expected fraction 0.40; binomial $p = 9.5 \times 10^{-5}$). The boundaries were predictions, not experimentally mapped attachment sites. Direct-repeat lengths at the boundaries did not differ from random windows (median ten base pairs in both groups, $p = 0.44$), and proximity to tRNA genes was not enriched (one of 29 versus eight of 58; $p = 0.26$).

Sequence redundancy limits independence. At 30% identity, the 183 proteins formed 23 clusters, with 127 proteins in the largest cluster. Selecting one representative per 50%-identity cluster left 79 loci. All 71 evaluable adjacent pairs remained on the same strand, and the median stored coordinate difference remained −3 bp. Ten of 79 loci were inside predicted proviruses (coverage-based binomial $p = 1.7 \times 10^{-6}$). Boundary proximity was less secure: seven of ten lay in the outer fifth, with $p = 0.055$ (Figure S4B). Thus, provirus association persists under this redundancy reduction, whereas preferential end localization remains provisional. Sequence clustering reduces redundancy without guaranteeing phylogenetic independence.

Interaction predictions do not establish a directionality mechanism

The campaign's paired-alignment analysis comprised 12,279 sequence pairs at an effective depth of approximately 1,845. Its strongest interprotein APC-corrected mutual-information score was 0.148, compared with within-protein maxima of 0.388 and 0.327 (Figure S4D). These statistics should not be equated with experimentally verified contacts or a fitted direct-coupling model [32]. Similarly, co-folding with 175 genomically paired alignments gave an interface predicted TM score (ipTM) of 0.538, compared with 0.543 after shuffling pair identities. Pairing therefore did not improve this score in the reported comparison (Figure S3D).

Boltz-2 models provided an additional, inconclusive line of evidence [33]. DUF2924 scored approximately 0.18–0.21 with candidate DNA substrates and 0.101 with its recombinase. However, the recombinase–DNA model and recombinase homodimer also had low scores (0.186 and 0.125), whereas LacI with its operator scored 0.905 (Figure 3I,J,L). These controls show that the workflow did not reliably recover the tested recombinase complexes under the chosen inputs. They do not establish that structure prediction has no value for the family, and low scores cannot exclude a native interaction. Structural prediction has identified experimentally functional LSR–RDF pairs elsewhere, emphasizing the need for system-specific calibration [21, 34].

The standalone DUF2924 collection was strongly basic, with a median predicted isoelectric point of 10.37 (interquartile range 9.92–10.87). These physicochemical features and the operon-level association motivate comparison with RDFs, but neither is diagnostic. The original five-factor comparator is not an exhaustive catalog, and its supplied length table misidentifies Bxb1 gp47 as a 56-residue protein, whereas the primary report describes an approximately 28-kDa protein [16]. We therefore do not retain the original claim that DUF2924 is 1.9-fold larger than characterized RDFs, nor a class-wide claim that RDFs share a uniform genomic arrangement [19, 21].

We propose OPAL, *operon-coupled accessory of large serine recombinases*, as a provisional designation for DUF2924 in this context, retaining PF11149 and DUF2924 for traceability and acknowledging its partial correspondence with RAMA. The term records a candidate accessory relationship, not a demonstrated RDF activity or an official database rename. The `TGA` stop within the recurrent `ATGA` junction also motivates the name, although stop and start remain distinct, overlapping codons.

Discussion

This study identifies a strong association between DUF2924 and large serine recombination modules. The principal evidence is genomic: a population-wide odds ratio of 58.2 relative to tyrosine-integrase loci, predominantly immediate adjacency, shared strand orientation and frequent CDS overlap. Junction-spanning RNA-seq reads support cotranscription at one representative locus, and localization within predicted proviruses links a subset of these modules to mobile DNA. Together, these observations justify prioritizing DUF2924/OPAL as a candidate accessory family. They do not establish its substrate, physical partner or effect on recombination.

The statistical reformulation was central to this interpretation. The initial case–background screen was useful for nomination but confounded by protein length and contig composition. Counting DUF2924 around all detected recombinases replaced that sampled background with explicit serine- and tyrosine-recombinase populations. This changed the estimand and made the association easier to interpret. It did not eliminate every source of bias: the two populations can differ in taxonomy, genome quality, element composition and annotation detectability. Genome-matched comparisons and sequence-cluster sensitivity analyses provided complementary checks, but further work should use taxonomically stratified models and genome- or clade-level resampling. Nominal locus-level significance should not substitute for these controls.

The campaign also illustrates the value of preserving unsuccessful analytical paths. The original programmable-system objective was not met. A fusion interpretation was revised after full-sequence inspection, a host-associated enrichment disappeared when the case definition changed, and motif-density and co-folding analyses remained inconclusive. Recording these revisions makes the basis of the final claim inspectable. However, a single campaign cannot establish that Luria outperforms human investigators or conventional pipelines, and an internal manuscript gate is not an independent validation study. Prospective comparisons across multiple campaigns, shared datasets and predefined evaluation criteria would be required to assess scientific productivity, reliability and cost relative to alternative workflows [10, 11, 12].

OPAL must also be positioned against existing biological knowledge. RAMA was proposed as a modified-adenine-recognition domain, and RAMA-containing MPND has been investigated biochemically and structurally [23, 26, 27]. Prior *Wolbachia* studies described DUF2924 or related annotations in mobile-element neighborhoods, including proximity to recombinases [28, 24, 29]. The contribution here is therefore the quantified scale and specificity of the association, together with its gene organization and explicit tests of alternative interpretations. A nomenclature proposal should improve retrieval across PF11149, DUF2924, RAMA and OPAL rather than obscure these connections or imply that all RAMA proteins share the same activity.

Several models remain compatible with the data. OPAL could influence recombinase activity through protein binding, recognize DNA or another nucleic acid, or participate in an element-associated process whose regulation benefits from shared expression. RDFs provide a plausible comparison because their interaction with LSRs can control recombination directionality [16, 17, 18, 19, 20]. Yet RDF sequence and structural diversity, including recently validated families, make charge and gene proximity inadequate functional classifiers [21]. Notably, known RDFs lack synteny with the integrase gene[35], whereas the OPAL association is almost entirely syntenic. OPAL may therefore represent a distinct, syntenic class of accessory factors, or its adjacency may reflect operon-level coupling rather than direct integrase binding; the proposed functional assays distinguish these possibilities. The present negative prediction controls neither support a particular mechanism nor exclude a genuine interaction. Likewise, GANTC motif density is not a methylation assay: modification-sensitive binding would require substrates whose sequence and methylation state are varied independently.

The strongest next experiments are direct functional tests. A defined LSR should be assayed for both integration and excision with and without its native OPAL partner, using cognate attachment sites and matched expression controls. Purified-protein assays should test recombinase binding and binding to unmethylated and methylated nucleic acids. Conserved C-terminal residues and the basic surface provide hypotheses for mutational analysis, subject to controls for folding and stability. Reporter constructs that alter the upstream start or stop codon while preserving relevant RNA features would test translational coupling directly [8, 30]. These experiments would distinguish an operon-associated candidate from a mechanistically defined recombination accessory factor.

The available sequences provide a substantial starting collection for these tests, but sequence counts should not be interpreted as counts of independently functioning tools. OPAL currently names a reproducible genomic relationship. Establishing whether that relationship governs mobile-element integration or excision requires experimental evidence

beyond association and prediction.

STAR Methods

Resource availability

Lead contact. Requests for resources and further information should be directed to the lead contact, Zehua Zeng (steorra@omicos.cn).

Materials availability. This computational study generated no new physical materials.

Data and code availability. The project repository for the campaign ledger, source tables and three independent scripts for recomputing the reported statistics is https://github.com/omicverse/luria-opal. Intermediate working files are archived on Zenodo at https://doi.org/10.5281/zenodo.23251666; the archive contains approximately 1.24 GB of uncompressed working files. Public sequence resources and accession identifiers supplied for this study are listed in the key resources table. Additional information required to reanalyze the reported results is available from the lead contact upon request.

Key resources table

Resource	Source	Version or identifier
Deposited data and sequence resources		
GTDB representative genomes	GTDB [1]	Release r232; 199,923 genomes
Viral RefSeq and phage genomes	RefSeq [2]; campaign inputs	Release dates and complete accession manifest to be supplied
Pfam profiles	Pfam [3]	v38.2;`Pfam-A.hmm`; 30,134 models; DUF2924: PF11149
R. capsulatus SB1003 reference	NCBI RefSeq	`NC_014034.1`
R. capsulatus SB1003 RNA-seq	NCBI Sequence Read Archive	`SRR12463562`; `SRR12463566`
Protein structure search database	PDB100	Local database built in July 2025
Campaign ledger, source tables and three recomputation scripts	This study; GitHub	`omicverse/luria-opal`
Intermediate working files	This study; Zenodo	doi:10.5281/zenodo.23251666; 1.24 GB uncompressed
Software and algorithms		
Luria orchestration harness	This study	Code revision and configuration to be supplied
Language model	DeepSeek	DeepSeek-4-Pro; deployment identifier to be supplied
Prodigal	Hyatt et al. [7]	v2.6.3
HMMER	Eddy [22]	v3.4; `hmmsearch --cut_ga`
BWA-MEM	Li [36]	v0.7.19-r1273
SAMtools	Danecek et al. [37]	v1.22.1
MAFFT	Katoh and Standley [38]	v7.526; `--auto`
FastTree	Price et al. [39]	v2.2.0; LG model; midpoint rooting
MMseqs2	Steinegger and Söding [40]	`easy-cluster`; version to be supplied
Boltz-2	Passaro et al. [33]	NVIDIA NIM service; deployment identifier to be supplied
Foldseek	van Kempen et al. [25]	Local build; commit `d5b1d3d`
geNomad	Camargo et al. [31]	v1.12.0; `find-proviruses`
ARAGORN	Laslett and Canback [41]	v1.2.41
Evo2	Brixi et al. [42]	40B; NVIDIA NIM endpoint `arc/evo2-40b`

Method details

The Luria harness and campaign accounting

Luria coordinated language-model instances through a shared, append-only ledger that recorded agent reasoning, analytical actions and results. A written mission brief initiated campaigns executed by a planning-and-analysis agent. An auditor checked reported quantities against the ledger, and a blinded experimental role reproduced measurements without access to their motivating interpretation. Rounds, declared beliefs, prospective falsification criteria, role dispatches and stopping decisions were appended to the ledger without rewriting earlier entries. At each outer-loop turn, the agent read the ledger to plan its continuation. A drafting agent assembled the manuscript from the ledger, and a manuscript gate required four tracked categories—missing evidence, unsupported statements, figure failures and structural failures—to contain no outstanding entries before accepting the draft. These workflow checks do not constitute independent validation of every result.

The campaign used DeepSeek-4-Pro and ran for 13 runtime turns. It opened 25 campaigns comprising 93 rounds, 18 declared beliefs containing 45 theories, seven prospective falsifiers, 76 role dispatches and 27 stops. Eleven roles were defined; the auditor was dispatched 63 times and the blinded experimental role 13 times. Reported token usage was

1,365 million tokens, including 10 million generated tokens. External direction was supplied at the start of each turn; autonomy therefore describes execution and revision within those directed intervals.

Counts in Figure 1A,C include only events attributable to a campaign identifier. Events without such an identifier were excluded, yielding 93 rounds and 76 dispatches compared with 96 rounds and 79 dispatches in the raw ledger. This distinction was maintained when summarizing campaign activity.

Sequence data and discovery entry points

The survey covered 618,638,921 proteins from GTDB representative genomes (release r232; 199,923 genomes and 32,515,015 contigs), viral RefSeq and phage-genome resources [1, 2]. Protein-coding genes were predicted with Prodigal v2.6.3 [7]. Contig identifiers, start and end coordinates, and strand were parsed from Prodigal FASTA headers, without requiring a separate annotation file.

Five complementary discovery entry points interrogated gene context, noncoding RNA, domain architecture, predicted structure and DNA repeat arrays. The context-first branch recovered host Holliday-junction resolvases. The structure-first branch modeled eight orphan folds without recovering a match, and the repeat-array branch promoted no candidate from 500 bacterial genomes and 5,000 phage genomes. These outcomes were retained as negative search results rather than excluded from the campaign record.

Pfam enrichment near integrase loci

The exploratory case collection comprised the host-Xer-cleaned orphan set, `clean_case_v2`, containing 79,415 proteins on 72,086 contigs. The background collection contained 102,336 proteins on 97,675 contigs. Samples of 10,000 proteins were drawn at random from each collection using seed 42. Domains were annotated with HMMER v3.4 `hmmsearch` against `Pfam-A.hmm` (30,134 models), using curated gathering thresholds and the options `--cpu 4 --cut_ga --domtblout` [3, 22].

Two-sided Fisher's exact tests compared domain-positive and domain-negative counts between the case and background samples. Benjamini–Hochberg adjustment was applied across 4,857 tested families [9], of which 32 reached $q < 0.05$. Zero-background families were displayed in the campaign summaries using a 0.5 pseudocount and a $\geq$ marker. That display convention is not interpreted here as a confidence bound or as a fully specified Haldane–Anscombe odds-ratio correction; the exact calculation requires confirmation from the analysis script.

An earlier case definition included host Xer-associated loci. Reclassification removed 20,585 of 100,000 candidate loci, leaving 79,415. The reported *pyrG* association decreased from 5.67 to 0.26, whereas the *guaA* association increased from 3.44 to 4.28. These comparisons illustrate the effect of case definition; they were not used as independent evidence for DUF2924 function.

Controls for the enrichment screen

Housekeeping-domain, length and genomic-context controls assessed sensitivity to the background population. All eleven housekeeping domains had odds ratios below 1.0, with a median of 0.566. Case and background proteins had median lengths of 211 and 273 residues, respectively; 36% and 23% were classified as short in the campaign summaries.

Repeating the family tests among proteins longer than 400 residues yielded 16 families passing the campaign's length-stratified gate, eleven failing it and five with an empty stratum. This analysis could not resolve length confounding for families lacking eligible proteins. The parent collections shared 5,757 contigs, equivalent to 5.9% of background contigs. Restricting the comparison to shared contigs yielded 30 DUF2924-positive proteins among 8,491 case proteins and none among 7,558 background proteins (reported $p = 5 \times 10^{-9}$). Matching on shared contigs reduced one source of confounding but did not establish equivalence of the sampled protein populations.

Whole-proteome association with recombinases

PF11149 was scanned across the full protein collection using the same `--cut_ga` threshold as the exploratory screen, identifying 15,266 unique proteins on 13,285 contigs. Each serine recombinase locus (n = 524,414) and tyrosine integrase locus (n = 1,403,550) was evaluated for the presence of at least one DUF2924 protein within five genes.

Positive loci numbered 14,088 and 665, respectively. For DUF2924-positive counts a and c among totals n_s and n_t, the odds ratio and prevalence ratio were calculated as

$$\mathrm{OR} = \frac{a/(n_s - a)}{c/(n_t - c)}, \qquad \mathrm{PR} = \frac{a/n_s}{c/n_t}.$$

These quantities were 58.2 and 56.7, respectively. The two-sided Fisher's exact-test probability underflowed double precision and is therefore not reported as an exact zero.

Gene-distance distributions covered distances zero through five in either direction. For DUF2924-centered pair-distance analyses, only the nearer recombinase was retained when recombinases occurred on both sides. This nearest-partner rule makes retained DUF2924 loci and pairs one-to-one within that analysis; it does not imply that distinct recombinase-centered windows are independent or that the two analyses share a denominator.

Locus reconstruction and operon geometry

The reference set comprised 183 DUF2924 loci on 181 reference contigs totaling 168.6 Mb. Coordinates were parsed from Prodigal headers. Of these loci, 168 had an immediately adjacent partner carrying a Resolvase domain and formed the operon-geometry set; 153 partners also carried a Recombinase domain. Strand and intergenic spacing were derived from the inclusive gene coordinates. Stored endpoint differences outside −60 to 600 bp were excluded as potential assembly or gene-calling artifacts before summarizing the spacing distribution.

For left and right genes, the stored spacing was $g = s_{\mathrm{right}} - e_{\mathrm{left}}$. The number of intervening nucleotides is $g - 1$, and overlapping genes have overlap length $1 - g$ when $g \leq 0$. Thus, $g = -3$ corresponds to a four-nucleotide overlap, and $g = 0$ denotes a one-nucleotide overlap. The 144 overlapping pairs were classified by true overlap length modulo three and compared with a uniform distribution using a chi-square test. Because start/stop codon constraints can themselves produce a nonuniform distribution, this test describes overlap geometry without establishing a mechanism of translational coupling [8, 30].

RNA-seq alignment and junction-spanning reads

Two public *Rhodobacter capsulatus* SB1003 datasets, `SRR12463562` and `SRR12463566`, comprised 29.6 million reads in total. Reads were aligned as single-end sequences to `NC_014034.1` using BWA-MEM v0.7.19-r1273 [36], after indexing the reference with `bwa index ref.fna`; alignment used `bwa mem -t 8`. Alignments were sorted and indexed with SAMtools v1.22.1 [37]. Per-base depth, including zero-coverage positions, was obtained with `samtools depth -a` over `NC_014034.1:2908000--2918000`.

At this locus, `NC_014034.1_2684` encodes a 441-residue recombinase with Recombinase, Resolvase and Zn_ribbon_recom domains. Its neighbor, `NC_014034.1_2685`, encodes a 152-residue DUF2924 protein. Their coding sequences overlap by four nucleotides (`ATGA`) on the minus strand. An alignment was counted as junction-spanning when it covered the complete junction window. Eighteen and five reads, respectively, satisfied this criterion, giving 23 across both runs. These observations support cotranscription at this locus; they do not by themselves demonstrate a shared promoter or translational dependence.

Phylogenetic analysis

Partner recombinase catalytic domains were extracted using alignment coordinates from the `hmmsearch` domain table and aligned with MAFFT v7.526 using `--auto` [38]. A tree was inferred with FastTree v2.2.0 under the LG model and rooted at its midpoint [39]. Label clustering was summarized as the number of adjacent tip pairs sharing the same DUF2924-neighbor label and assessed against 5,000 label permutations. The same alignment and tree-inference workflow was applied to the 183 DUF2924 domains shown in Figure S3A.

Sequence clustering and redundancy sensitivity

Sequences were clustered with MMseqs2 `easy-cluster` [40], using `--min-seq-id 0.30 -c 0.5 --cov-mode 1`. The cluster-based reanalysis of the screening collection yielded 84,591 clusters. The 183 reference DUF2924 proteins were additionally clustered at 30%, 50% and 80% identity with the same coverage settings. At 30% identity, 23 clusters

were obtained, the largest containing 127 sequences; at 50% identity, 79 clusters were obtained. Locus-level statistics summarized in Figure S4B were recomputed using one representative per 50%-identity cluster. This sensitivity analysis reduces sequence redundancy without eliminating phylogenetic dependence. Genome provenance was tabulated from assembly-source labels associated with the 181 reference sequences.

Sequence conservation

The 183 DUF2924 domain sequences were aligned and conservation was summarized by alignment column. Of 194 columns, 134 were occupied in more than half of the sequences and defined the core alignment. Conservation in columns 171–190 was compared with the remaining core columns. These are alignment coordinates, not residue positions shared by every protein. PF11149 profile information content was extracted from the Pfam HMM, and its correspondence with column-wise sequence conservation was assessed by Spearman correlation.

Structure prediction, fold searches and charge

Twenty-four DUF2924 proteins spanning the sampled family were modeled with Boltz-2 through NVIDIA NIM, using multiple sequence alignments derived from the 183 family members [33]. Reported model-confidence values ranged from 0.51 to 0.66. Models were searched against PDB100 using a local Foldseek build (commit `d5b1d3d`) and a database built in July 2025 [25]. The supplied search specification used `easy-search`, `-e 1e-3`, `--max-seqs 50`, and the output fields `query,target,fident,alntmscore,evalue,prob`.

No model had a reported match with $E < 10^{-3}$. The exploratory outputs nevertheless listed a best match of $E = 0.074$ and MPND matches for three models with Foldseek probabilities of 0.941, 0.986 and 0.382. These above-threshold hits cannot be attributed to the stated reporting cutoff without additional search records. These matches are treated as hypothesis-generating comparisons rather than confident fold assignments.

Sequence net charge was approximated as $100(N_{\mathrm{Arg}} + N_{\mathrm{Lys}} - N_{\mathrm{Asp}} - N_{\mathrm{Glu}})/L$, where L is sequence length. Distributions were compared for DUF2924 proteins ($n = 183$), length-matched neighborhood proteins ($n = 325$), all neighborhood proteins ($n = 1{,}572$) and partner serine recombinases ($n = 238$) using Mann–Whitney tests. This composition-based index does not represent a pH-dependent electrostatic calculation. Surface basic/acidic area summaries were obtained from six rotational views of one model.

Complex prediction and interprotein sequence covariance

Protein and nucleic-acid complexes were predicted with Boltz-2 through NVIDIA NIM, using five diffusion samples per complex [33]. Interface predicted TM scores (ipTM) were summarized as means across samples. Paired multiple sequence alignments linked each of 175 DUF2924 proteins to a recombinase encoded on the same contig. Shuffled controls permuted these pairings while preserving alignment depth; unpaired predictions used server-generated alignments.

Controls included the partner recombinase with provirus-edge DNA, the recombinase homodimer, a large serine recombinase from a locus lacking DUF2924, a tyrosine integrase, shuffled nucleic-acid sequences, and LacI with its cognate or scrambled operator. These controls assess prediction behavior under the specified settings and do not establish physical binding.

A separate paired-sequence analysis used 12,279 sequence pairs with a reported effective depth of 1,845. Average-product-corrected (APC-corrected) mutual information was used to compare the strongest interprotein covariance with the strongest intraprotein covariance in each partner [43]. Because no fitted direct-coupling model was specified, this analysis is described as sequence covariance rather than direct-coupling analysis.

Mobile elements, boundary geometry and attachment-site searches

Proviruses were predicted with the `find-proviruses` module of geNomad v1.12.0 across the 181 reference contigs (168.6 Mb) [31]. The analysis identified 87 elements covering 1.8% of total contig length. For each of 29 DUF2924 loci within a predicted element, position was normalized by element length and distance to the nearer boundary was calculated. The proportion within the outer fifth at either end was assessed by a binomial test with a null probability of

0.40. Normalized nearest-boundary distances were also compared with Uniform$(0, 0.5)$ using a Kolmogorov–Smirnov test. These idealized null models do not account for nonuniform gene density or dependence among related elements.

Candidate attachment sites were sought as the longest exact direct repeat shared between the two ends of each analyzed element ($n = 63$); repeat lengths were compared with random window pairs from the same contig. ARAGORN v1.2.41 identified tRNA genes in windows extending 500 bp on either side of each element boundary, with equal-length control windows located 20 kb away on the same contig [41]. Junction-sequence constraint was assessed with Evo2 40B through the NVIDIA NIM endpoint `arc/evo2-40b`, comparing reconstruction identity at the junction with random windows ($n = 80$) [42].

Methyltransferase neighborhoods and the GANTC test

Neighborhoods were screened with 38 methylation-related HMMs. Counts were then restricted to nine DNA methylation and restriction–modification domain families within five genes of DUF2924: N6_N4_Mtase, N6_Mtase, DNA_methylase, Methylase_S, MmeI_Mtase, Dam, EcoRI, N6-adenine and HTH_ParB_Mtase. Using all 38 models yielded frequencies of 20.8% versus 3.5%, preserving the direction of the association under the broader annotation set.

Two control sets were evaluated: integrase-neighborhood orphan proteins matched for integrase adjacency ($n = 200$) and random recombinase loci ($n = 200$; 60 serine and 140 tyrosine recombinases). The integrase-adjacent orphan set was the principal matched comparison; the mixed random-recombinase set addressed a different background. Candidate recognition specificities were tabulated from Pfam-family annotations and were not treated as experimentally measured substrate preferences.

The GANTC hypothesis test was recorded as a prospective falsification criterion before measurement. Groups were matched by element class and host clade: group A contained alphaproteobacterial serine recombinase loci with a DUF2924 neighbor ($n = 7{,}758$), and group B contained corresponding loci without one ($n = 15{,}000$). GANTC and GATC motif densities in windows around each locus were compared using standardized effect sizes. GATC served as a control motif for the proposed CcrM/GANTC mechanism. Sequence motifs alone do not measure DNA methylation.

Comparison with characterized directionality factors

Five experimentally characterized recombination directionality factors were selected from the literature: Bxb1 gp47, ϕC31 gp3, ϕRv1 Rv1584c, TP901-1 ORF7 and A118 Gp44 [16, 17, 18, 20, 19]. Protein length and predicted isoelectric point were compared with DUF2924, and gene order relative to the partner recombinase was tabulated from Prodigal coordinates. The five proteins constitute a historical comparator panel rather than an exhaustive catalog of known directionality factors [21].

Quantification and statistical analysis

Fisher's exact tests assessed family enrichment and categorical proportions; family-screen probabilities were adjusted with the Benjamini–Hochberg procedure across 4,857 families. Continuous distributions were compared using Mann–Whitney tests, overlap-length residue classes using a chi-square test, and element-boundary proximity using binomial and Kolmogorov–Smirnov tests. Tree-label clustering was assessed with 5,000 permutations and summarized as a z score relative to the permutation distribution. Spearman correlation assessed agreement between column-wise conservation and profile information content.

The supplied protocol specifies two-sided tests where applicable. Sample sizes and denominators are stated for each analysis; protein-, locus-, contig- and element-level observations are not interchangeable. Control matching was analysis-specific, as described above, rather than assumed to be identical across all comparisons. Sequence-cluster sensitivity analyses were used to assess redundancy, but did not remove all phylogenetic or genomic dependence. Nominal probabilities from simplified placement or phase null models were interpreted alongside effect sizes and the constraints of those models.

Author contributions, funding and competing interests

These declarations require author-supplied information before submission. Contributions, funding sources and competing-interest statements have not been inferred from author order or affiliation.

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Main figures

The figure artwork is reproduced from the supplied manuscript. Corrections to coordinate conventions and comparator data are identified with the relevant figure; the revised text and legends govern their interpretation.

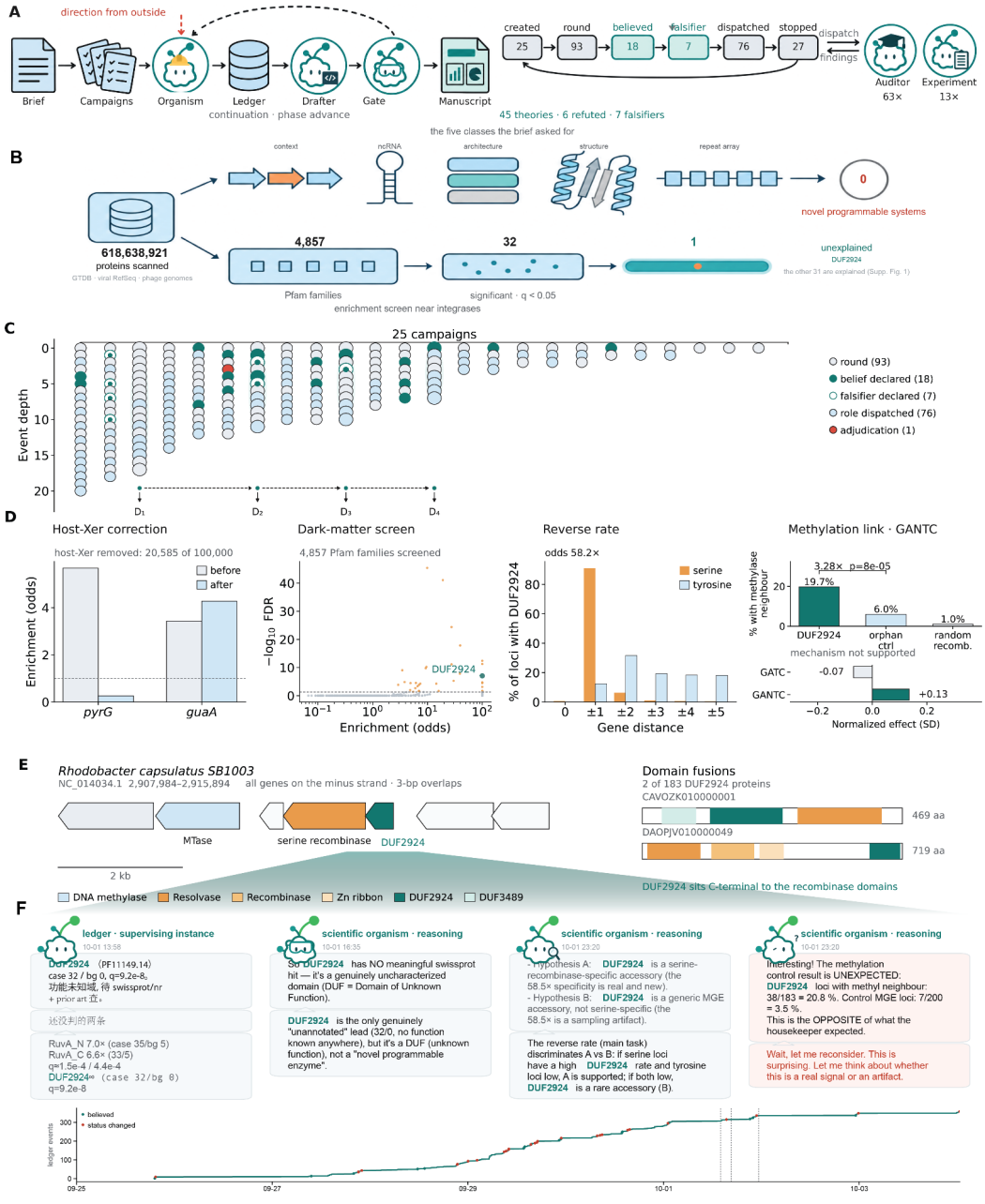


Figure 1: An autonomous genome-mining campaign nominates DUF2924.

Figure 1 legend

A, Luria workflow and ledger event counts. Campaigns record rounds, beliefs, falsifiers, dispatches and stops; a drafting step and manuscript gate use the resulting record. **B**, Five complementary search strategies applied in the campaign, none of which recovered a new programmable system under the stated criteria. The secondary enrichment screen identified 32 families at reported $q < 0.05$, with DUF2924 remaining unexplained after annotation review. **C**, Events within each campaign, with D1–D4 indicating the principal campaigns leading to DUF2924. **D**, Changes in the inferred associations: removal of host Xer sites reduced the *pyrG* signal; Pfam enrichment nominated DUF2924; the recombinase-centered comparison yielded an odds ratio of 58.2; and the methyltransferase-neighborhood association did not translate into a substantial GANTC-density difference. **E**, Representative gene organization and the two observed recombinase–DUF2924 fusions. A stored coordinate difference of −3 denotes a four-nucleotide overlap for inclusive coordinates. **F**, Excerpts from the campaign record, preserved as historical statements rather than endorsed final interpretations, and the cumulative event trajectory. References to an uncharacterized family in these excerpts precede retrieval of the RAMA and *Wolbachia* literature [23, 24, 28, 29]. See also Figures S1 and S2.

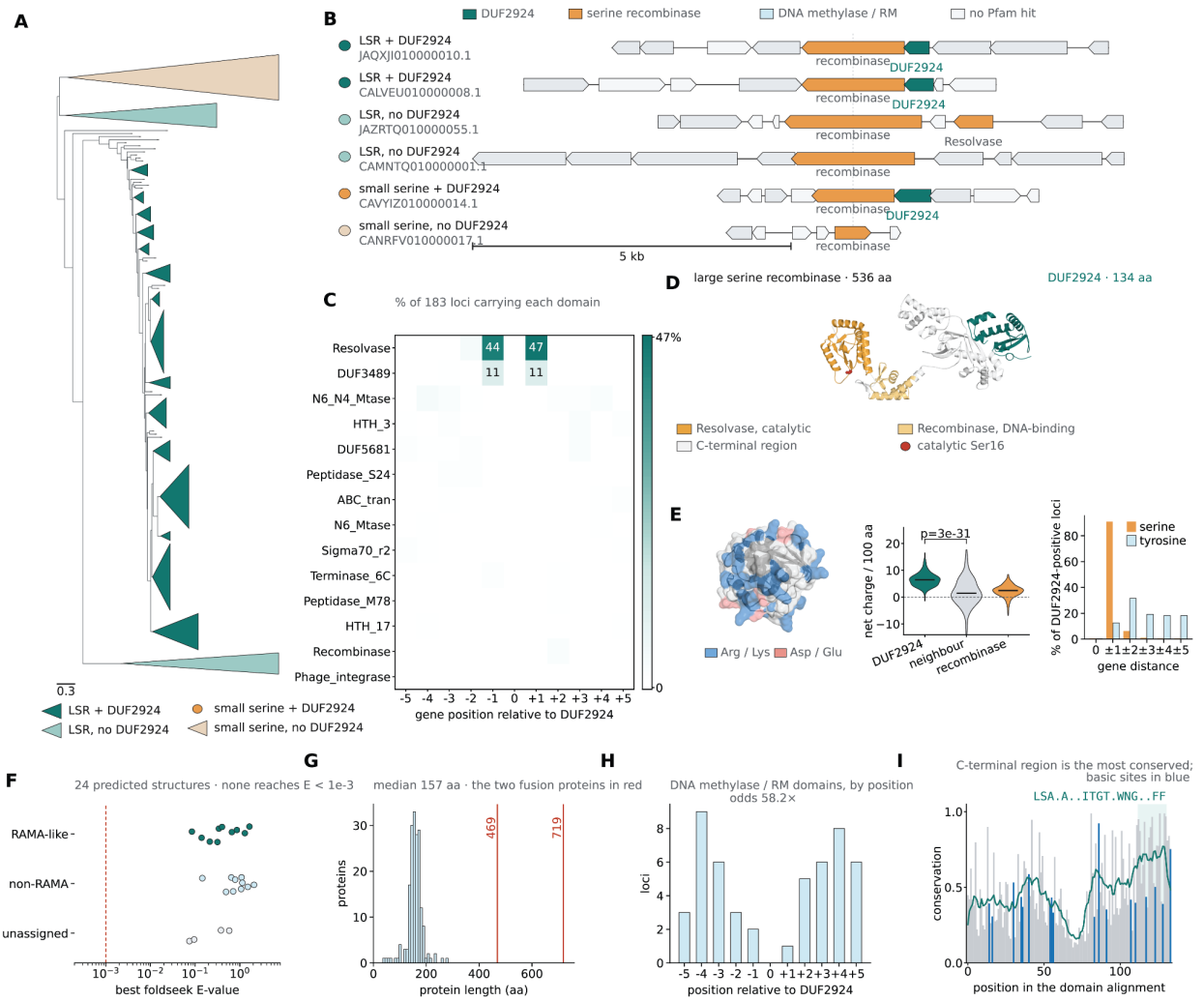


Figure 2: DUF2924 partners are predominantly large serine recombinases, and DUF2924 proteins are small and basic.

Figure 2 legend

A, Midpoint-rooted phylogeny of sampled recombinase catalytic domains, grouped by DUF2924 association and domain architecture. Same-label tip adjacency is descriptive and is sensitive to branch ordering. **B**, Representative coordinate-resolved neighborhoods. **C**, Pfam-domain prevalence by gene position across 183 DUF2924 loci. The strongest recurrent annotations occupy the two positions flanking DUF2924. **D**, Representative predicted structures, including the annotated catalytic serine in the recombinase. **E**, Predicted DUF2924 surface charge, sequence-derived net charge relative to comparison proteins and recombinase-centered gene-distance distributions. The 91.0% immediate-adjacency fraction uses the serine-recombinase-centered denominator. **F**, Best Foldseek match for each of 24 predicted DUF2924 structures; no model reached the campaign threshold of $E < 10^{-3}$ [25]. **G**, Protein lengths, including two longer fusion proteins. **H**, Positions of methyltransferase and restriction-modification annotations. **I**, Conservation over the aligned core, highlighting the C-terminal region. Positively charged surfaces and weak structural matches motivate biochemical tests but do not establish nucleic-acid recognition or modified-base specificity. See also Figure S3.



Figure 3: Gene organization, transcript evidence and predicted provirus localization support an operon-level association.

Figure 3 legend

A, Data collections used by the panels. The 181 genomic inputs comprise the analyzed reference contigs and should not all be assumed to be complete genomes. **B**, A representative DUF2924–recombinase locus in a predicted provirus of *D. ovata*. **C**, Nearest-serine-recombinase distance among 13,494 pairable DUF2924-centered loci; 12,790 (94.8%) are immediately adjacent. **D**, Representative `ATGA` junctions and spacing for 168 same-strand adjacent pairs. The plotted spacing g is an inclusive-coordinate endpoint difference. Thus $g = -3$, found at 89 loci, denotes a four-nucleotide CDS overlap; $g = 0$, found at 17 loci, denotes a one-nucleotide overlap. In total, 144/168 pairs overlap, rather than the 76% shown for the strictly negative subset. The displayed modulo-three bins are based on $-g$; true overlap lengths are $1 - g$, shifting the classes by one. The reported chi-squared statistic describes departure from a uniform distribution, which is not a sufficient biological null for codon-constrained overlaps. **E**, RNA-seq coverage at the *R. capsulatus* SB1003 locus; 18/41 and 5/14 junction-window reads span the junction in two runs, supporting cotranscription. **F**, Partner length and domain architecture. Nonsignificant tests do not demonstrate equivalence. **G**, Twenty-nine of 183 loci fall within predicted proviruses occupying 1.80% of analyzed sequence. The 8.8-fold value is a locus-fraction/sequence-coverage ratio. The original Fisher label is not used for inferential support; a coverage-based binomial null has different assumptions and gives $p \approx 7.0 \times 10^{-19}$. **H**, Positions within predicted proviruses; boundaries are not mapped attachment sites. Boundary proximity loses conventional significance after sequence-cluster reduction. **I,J**, Boltz-2 interface scores for candidate and control complexes [33]. Low scores for the tested recombinase controls prevent a negative binding inference for DUF2924. **K,M**, Original RDF comparison panels, retained for traceability but requiring revision. The primary Bxb1 report describes gp47 as approximately 28 kDa, invalidating the 56-aa entry [16]; the five-factor set is not exhaustive [21]. DUF2924's own predicted pI and gene-order measurements remain descriptive results. **L**, Representative predicted complexes, whose geometry is not experimental evidence of binding. See also Figure S4.

Supplementary figures

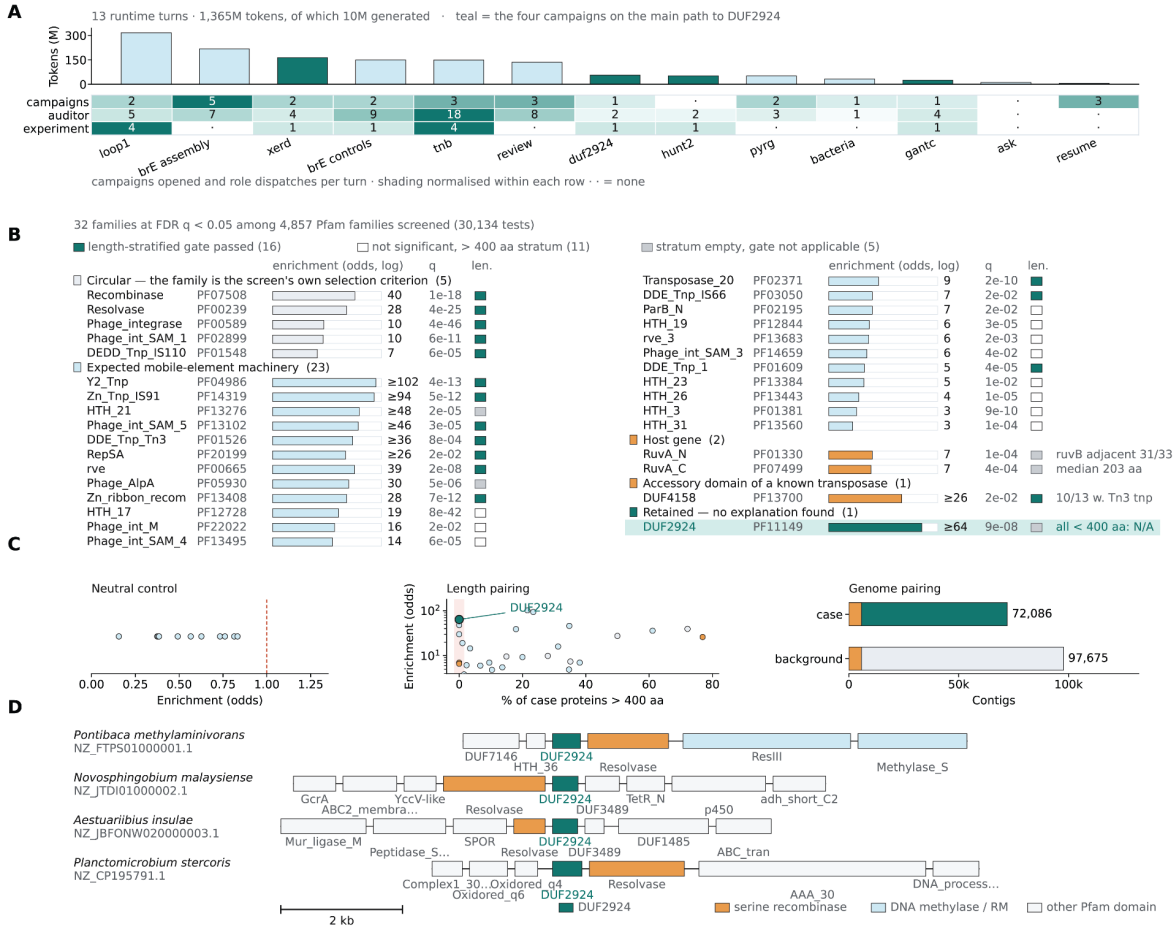


Figure S1: Screen outcomes, computational accounting and background limitations.

Figure S1 legend

A, Token use and campaign/role activity across 13 runtime turns. The reported 1,365 million tokens include ten million generated tokens; the remaining accounting categories require the original usage definitions. **B**, Thirty-two enriched Pfam families grouped by the interpretation assigned during the campaign. Zero-background categories use the stated correction for plotting finite odds. The length-restricted analysis includes only proteins longer than 400 residues and is unavailable for families without eligible observations. The main-text FDR denominator is 4,857 tested families; the original legend also mentions 30,134 tests, an inconsistency requiring verification from the full test table. **C**, Housekeeping-domain depletion, unequal protein lengths and limited contig overlap. These checks constrain interpretation of the nomination screen; they do not imply that the recombinase-centered follow-up is free of all confounding. **D**, Additional representative gene neighborhoods retained from the supplied artwork; gene arrows indicate the annotated local context.

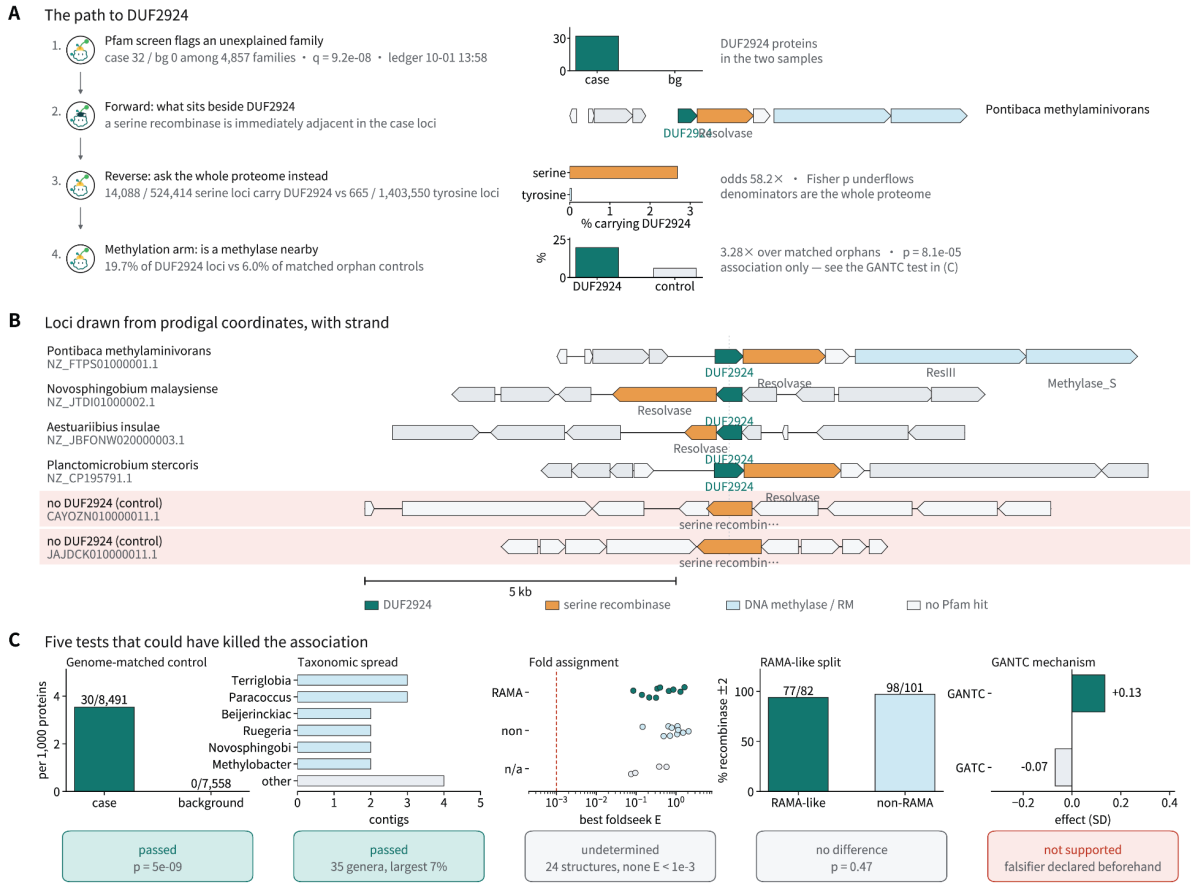


Figure S2: Follow-up comparisons supporting the DUF2924 association and limiting mechanistic inference.

Figure S2 legend

A, Candidate progression from enrichment to local gene organization and the whole-proteome recombinase-centered comparison. Methyltransferase co-occurrence is a separate neighborhood measurement. **B**, Coordinate-derived gene diagrams for DUF2924 loci and control recombinase loci. Control neighborhoods were scanned for methyltransferase models only; blank annotations should not be interpreted as absence of other domains. **C**, Shared-contig comparison (30/8,491 versus 0/7,558), reported taxonomic distribution, Foldseek searches, a RAMA-based split and motif-density comparisons. The RAMA split uses 82/183 annotated proteins and compares 77/82 with 98/101 recombinase-associated cases; this differs from the 80 and 172 counts used in Figure S3A and requires reconciliation of annotation thresholds and locus sets. GANTC/GATC density does not directly measure methylation or DNA binding [23, 27].

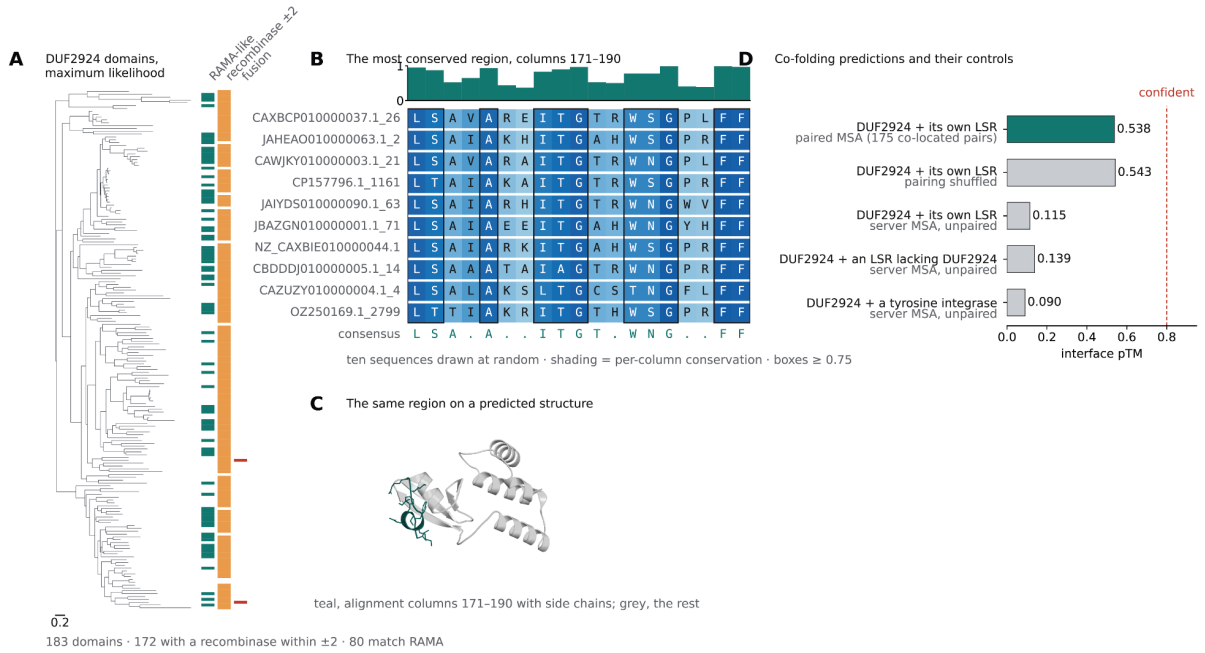


Figure S3: Sequence conservation, annotation distribution and co-folding controls.

Figure S3 legend

A, DUF2924 sequence tree with RAMA, recombinase-proximity and fusion annotations. This figure's supplied summary records 80 RAMA matches, 172 loci with a recombinase within two genes and two fusions; the corresponding Figure S2 summary uses different counts. **B**, Alignment columns 171–190 for ten sampled sequences, with the conserved C-terminal motif. Column numbers do not identify universally corresponding protein residue numbers without sequence-specific mapping. **C**, Predicted structural location of this segment. **D**, Co-folding scores from native paired alignments, shuffled pairs, unpaired alignments and alternative partners. The native-pair score of 0.538 does not exceed the shuffled score of 0.543. The dashed 0.8 reference line is a heuristic and is not a universally calibrated interaction threshold across prediction methods or molecule types.

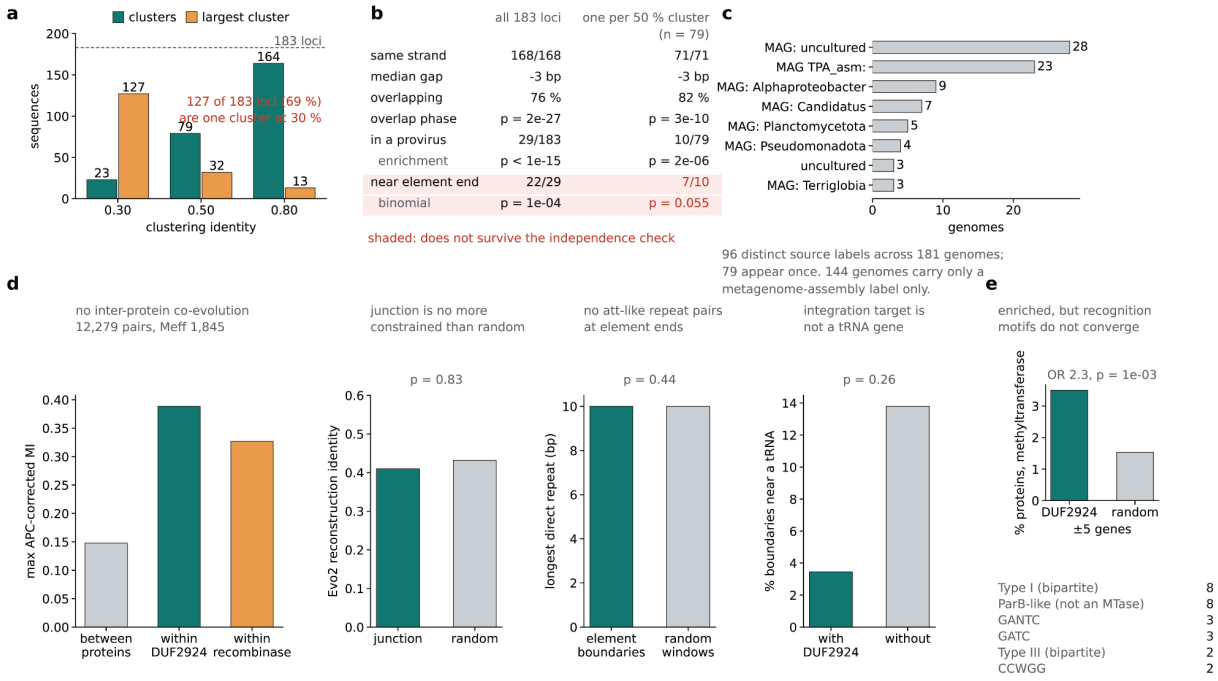


Figure S4: Sequence redundancy and additional inconclusive mechanistic tests.

Figure S4 legend

A, Clustering of 183 proteins at multiple sequence-identity thresholds. At 30% identity, the largest cluster contains 127 proteins. **B**, Full-set and 50%-identity-representative summaries. Shared strand and close spacing persist, and ten of 79 representatives lie within predicted proviruses. The seven-of-ten boundary-proximity result has a binomial $p = 0.055$. Overlap labels use the original stored coordinate convention; counts should be interpreted as explained in Figure 3D rather than as verified base-pair overlap statistics. **C**, Assembly source labels, which do not establish independent isolation. **D**, APC-corrected mutual-information summaries, an exploratory Evo 2 sequence reconstruction comparison [42], boundary direct-repeat lengths and tRNA proximity. The mutual-information measurement is not itself direct-coupling analysis. Evo 2 reconstruction identity was 0.410 at junction windows and 0.432 at random windows (reported $p = 0.83$, $n = 80$); this does not provide positive evidence for a special junction constraint. **E**, Methyltransferase-associated annotation counts across neighborhood proteins (55/1,572 versus 21/1,368; odds ratio approximately 2.3). These protein-level denominators differ from the 183-locus versus 200-locus comparison in the main text. ParB-like annotations included in the category are not all methyltransferases, limiting functional interpretation.